8.2 Bounding Without Solving

WORKSHEET

The question

(i) Let $y(x)$ be a solution of the differential equation $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2} + y^3 = 0$ with $y = 1$ and $\dfrac{\mathrm{d}y}{\mathrm{d}x} = 0$ at $x = 0$, and let

$$\mathrm{E}(x) = \left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2 + \tfrac{1}{2}y^4.$$

Show by differentiation that E is constant and deduce that $|y(x)| \leq 1$ for all x.

(ii) Let $v(x)$ be a solution of the differential equation $\dfrac{\mathrm{d}^2v}{\mathrm{d}x^2} + x\dfrac{\mathrm{d}v}{\mathrm{d}x} + \sinh v = 0$ with $v = \ln 3$ and $\dfrac{\mathrm{d}v}{\mathrm{d}x} = 0$ at $x = 0$, and let

$$\mathrm{E}(x) = \left(\frac{\mathrm{d}v}{\mathrm{d}x}\right)^2 + 2\cosh v.$$

Show that $\dfrac{\mathrm{dE}}{\mathrm{d}x} \leq 0$ for $x \geq 0$ and deduce that $\cosh v(x) \leq \frac{5}{3}$ for $x \geq 0$.

(iii) Let $w(x)$ be a solution of the differential equation

$$\frac{\mathrm{d}^2w}{\mathrm{d}x^2} + (5\cosh x - 4\sinh x - 3)\frac{\mathrm{d}w}{\mathrm{d}x} + (w\cosh w + 2\sinh w) = 0$$

with $\dfrac{\mathrm{d}w}{\mathrm{d}x} = \dfrac{1}{\sqrt{2}}$ and $w = 0$ at $x = 0$. Show that $\cosh w(x) \leq \frac{5}{4}$ for $x \geq 0$.

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

Ask yourself For part (i), think of y as the position of a particle and x as time. Then $y'' = -y^3$ is Newton's law with force $-y^3$. What are the kinetic and potential energies?

STAGE 2 Dissipation

Ask yourself Differentiate $\mathrm{E} = (v')^2 + 2\cosh v$. Use the differential equation to replace v''. What sign does the result have?

STAGE 3 Inventing the energy

Ask yourself If $\mathrm{E} = (w')^2 + F(w)$, then $\mathrm{E}' = 2w'w'' + F'(w)w'$. What should $F'(w)$ be, so that the equation can be used to replace w'' and leave only a multiple of $(w')^2$?

Pause and try Show that $5\cosh x - 4\sinh x - 3 \geq 0$ for all x. (Write it in terms of e^x and look for a perfect square.) Then finish the argument.

Try this next

1. **Simple harmonic motion.** For $y'' + y = 0$ with $y(0) = A$, $y'(0) = 0$, find the conserved energy and deduce $|y| \leq A$. Check against the exact solution.

2. **The pendulum.** The angle θ of a pendulum satisfies $\theta'' + \sin\theta = 0$ (in suitable units). Show that $(\theta')^2 - 2\cos\theta$ is constant. If the pendulum starts at rest at angle θ_0, where $|\theta_0| \leq \pi$, show that $|\theta|$ never exceeds $|\theta_0|$. What goes wrong if $|\theta_0| > \pi$?

3. **Damping.** For $y'' + 2ky' + y = 0$ with $k > 0$, show that $(y')^2 + y^2$ decreases. Does it necessarily tend to zero?

4. **An escape.** For $y'' - y^3 = 0$ with $y(0) = 1$, $y'(0) = 0$, find a conserved quantity and show that y is not bounded by it. What goes wrong?

Further reading

- **Steven Strogatz, *Nonlinear Dynamics and Chaos*** (CRC Press). Chapters 6 and 7 cover phase planes, conservative systems and Lyapunov functions.

- **D. W. Jordan & P. Smith, *Nonlinear Ordinary Differential Equations*** (Oxford University Press). A standard undergraduate text with many examples of energy methods.
- **Keith Devlin, *The Millennium Problems*** (Basic Books). Includes an accessible account of the Navier–Stokes problem.

The full guided solution, and what the question leads to at university, is in Section 8.2:
https://beyondstep.keithmcnulty.org/ch08/bounding-without-solving.html